

Operating System

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Quiz

Live Poll

About this Course

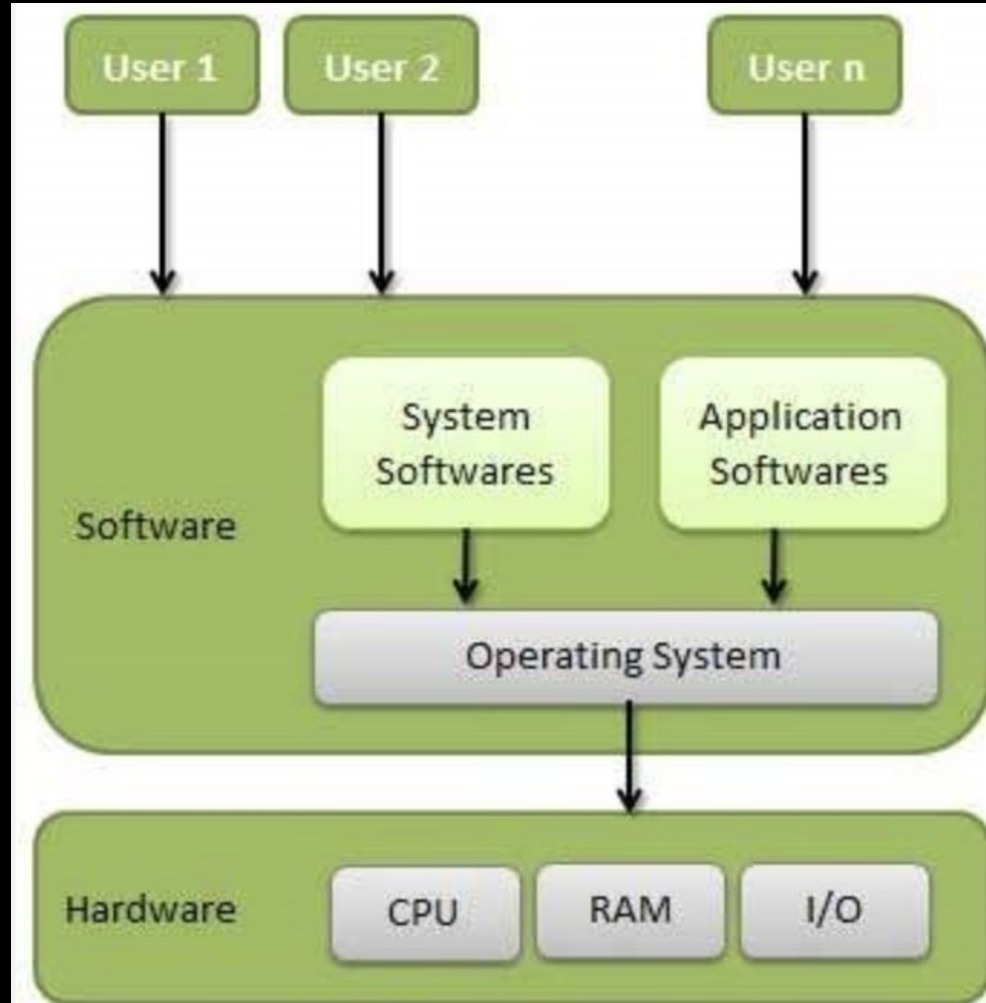
- OS Programming: to use OS
- OS Implementation: to write OS
- C Language
- Linux Commands

Core Components

- Syllabus Overview
 - *Three easy pieces*
 - Virtualization
 - Concurrency
 - Persistence

Why do we need OS?

- Why do we need Virtualization?



OS Fundamental Concepts

Virtualization

- CPU Virtualization
- Memory Virtualization

Concurrency

- threads

Persistence

- File System

OS: Three easy pieces

Virtualization (Process and Memory)

- 1.OS gives every process a feeling that it has own CPU
2. Every process feels that its enough memory

Concurrency

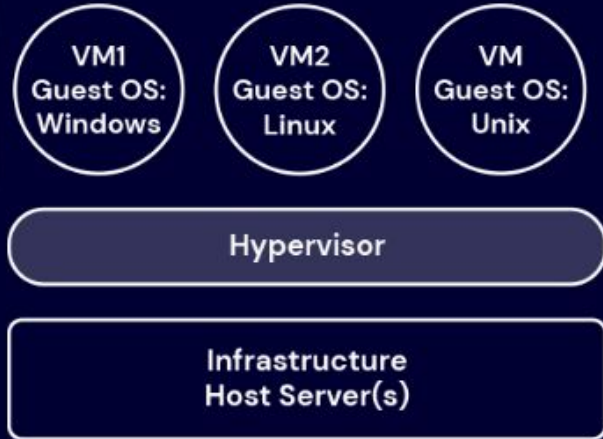
1. Multiple process can run at the same time without resulting in problems
2. OS Provides mechanisms to make them work together

Persistence

1. Disk is an I/O device. It needs to be managed and handled.
2. OS handles interactions with the disk and performs storage management

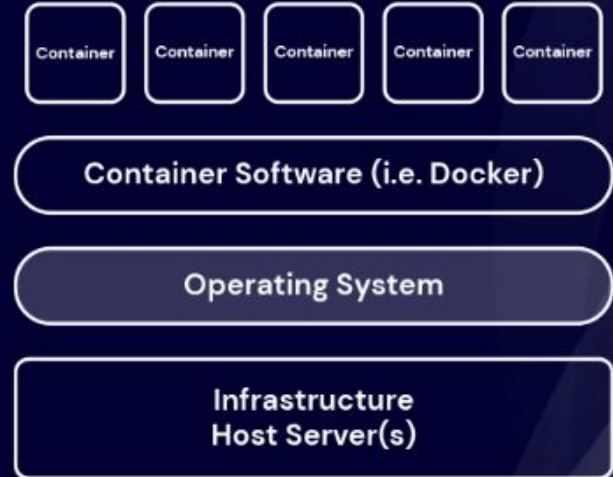
Virtual Machines vs Containers

Virtual Machines



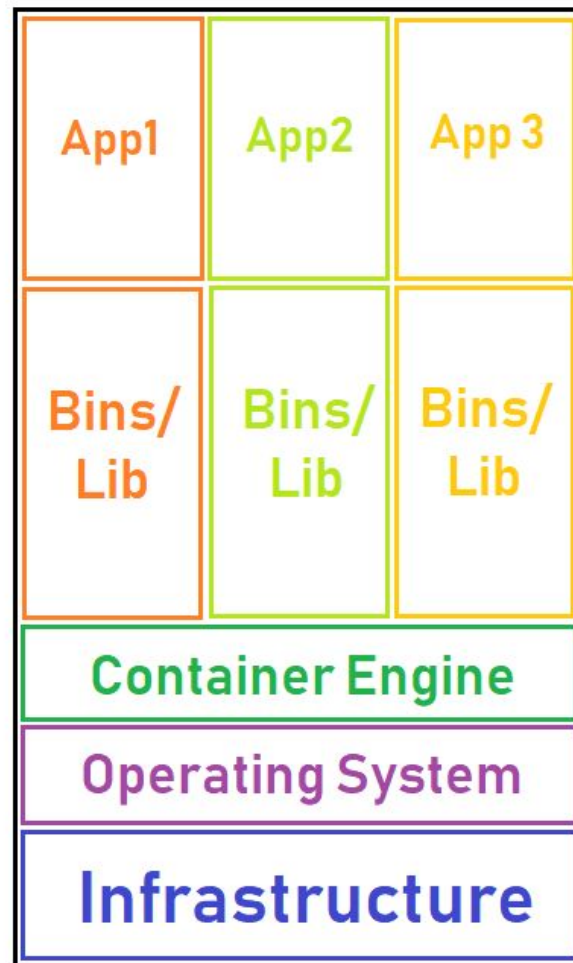
WHAT'S
the
DIFF?

Containers





Virtualization



Containerization